

Plan: Epinions AUC Lift Studies

(Meta-Plan: Candidate Levers & Sweep Order)

2026-05-10

1. Position and target

Reference	5-seed AUC
SGT (signed graph transformer, external)	0.941
SiKAN per-vertex baseline ($m=128$, prod config)	0.8464 ± 0.0106
Adaptive m_v ($c=1$) smoke (single-seed, abbrev. config)	0.7088 vs 0.6416 baseline (+0.067)
Adaptive m_v ($c=1$) 5-seed (production config, IN FLIGHT)	TBD (~ 3.8 h ETA)

The gap to SGT is 0.094 at the 5-seed level. If the adaptive m_v delta scales fully (+0.067), production-config AUC would land at ≈ 0.913 still 0.028 below SGT. We need additional *stackable* levers.

This plan is a meta-plan: a prioritised matrix of candidate levers and the experimental order to test them, gated by the in-flight 5-seed result. Each lever gets its own implementation plan only once it passes a single-seed smoke test on the production config.

2. Lever candidates

2.1 Quick reference table

Lever	Hypothesis	Predicted lift	Cost
A. Sub-unit c	Smoke trend monotonic; $c < 1$ extrapolates	+0.02 to +0.04	3 smoke runs
B. Adaptive + sparse attn	Synthetic-validated, untested on Epinions	+0.01 to +0.03	1 smoke (~ 3 h)
C. Davis pruner instead of balance	Different cycle pool; orthogonal to m_v	± 0.02	1 smoke
D. $m_{\max}$ sweep	Combine adaptive with smaller cap	+0.01	3 smoke runs
E. d_{hidden} wider	Slashdot null but Epinions might differ	-0.01 to +0.01	1 smoke
F. Walks-only	Drop cycles, walk-rich is Slashdot-good	-0.05 to +0.02	1 smoke
G. Different scorer	balance/sign_product_abs/low_root	± 0.01	4 smoke runs
H. More epochs	80 $\rightarrow$ 160	-0.01 to 0	1 smoke (2x)

2.2 Detailed lever rationale

A. Sub-unit c . The smoke c -sweep was monotonic decreasing in AUC as c grew from 1 to 16. Extrapolating the trend, $c < 1$ should lift further. At $c=0.5$, low-degree vertices ($\text{deg} \leq 1$) get $m_v = 1$; mid-degree ($\text{deg}=13$ avg on Epinions) get $m_v = 7$; hubs still saturate at 128. This sharpens the signal-density argument from the smoke report further. **Expected lift floor:** +0.02 over $c=1$ (i.e. on top of the in-flight 5-seed result). **Cost:** 3 smoke runs at $c \in \{0.25, 0.5, 0.75\}$ at the abbreviated config.

B. Adaptive m_v + sparse attention. Per `project_sparse_attention.synth_2026.05.10`, the sparse attention (`HSIKAN_SPARSE_ATTN_K=8`) is synthetic-validated but untested on real Epinions. The per-edge attention sparsifier is orthogonal to the cycle source change; combining both should be near-additive. **Expected lift:** +0.01 to +0.03. **Cost:** 1 smoke run.

C. Davis pruner instead of balance. Currently `HSIKAN_TOPK_PRUNER=balance` (Cartwright–Harary balanced cycles only). Davis weak balance keeps all cycles *except* all-negative triangles a larger cycle pool with different sign structure. Smoke would tell whether the larger pool informs HSiKAN better than the strict balanced subset. **Risk:** low (orthogonal to m_v). **Cost:** 1 smoke.

D. $m_{\max}$ sweep. We’ve held $m_{\max} = 128$ throughout. Combined with adaptive $c=0.5$, low-degree vertices get tiny caps already, so reducing $m_{\max}$ from 128 to 64 or 32 only affects hubs. If hubs are the over-saturation source, this should help further. **Cost:** 3 smoke runs at $m_{\max} \in \{32, 64, 128\}$.

E–H. Lower-priority; tested only after A–D land.

2.3 Stacking compatibility

Lever combination	Compatible?	Expected interaction
A (sub-unit c) + B (sparse attn)	yes	near-additive
A + C (Davis)	yes	substitution at the pruner level
A + D ($m_{\max}$)	yes	both shrink heaps
A + E ($d = 16$)	yes	$d = 16$ failed Slashdot separate test
B + C	yes	both cycle/attention level
B + D	yes	cycle pool + attention sparsity

The natural **stacking order** is $A \rightarrow B \rightarrow C$, choosing the best of each before stacking the next. Each stack-step requires its own 5-seed paired validation.

3. Sweep plan

3.1 Pre-5seed CPU-only studies (this plan)

While the in-flight 5-seed runs (3.8h), launch CPU-only cycle-distribution probes that don’t compete for the GPU:

- Extend `probe_per_vertex_thresholds.rs` with a `c` CLI argument; sweep $c \in \{0.25, 0.5, 1.0, 2.0, 4.0, 8.0\}$.
- Report per c : full-heap-rate, total cycle count, vertex coverage histogram, score distribution.
- Generate a matplotlib plot to visualize the trend.
- Predict the cycle-source landscape we’d train on at each c .

This gives us a data-grounded prediction for which c value to test next at training-time, ahead of running the smoke.

3.2 Post-5seed sweep order

Gated on the in-flight 5-seed result:

1. **If** adaptive $c=1$ at production passes the ± 0.005 AUC band over fixed- m baseline: confirmed; proceed to A–D.
2. Run lever A (sub-unit c) single-seed smokes at $c \in \{0.25, 0.5, 0.75\}$. Pick the best. Run 5-seed at the best c for promotion.
3. Run lever B (sparse attn) single-seed at the best c . Pick the best K_{attn} value. 5-seed at best.
4. Run lever C (Davis pruner) single-seed at the best c , best K_{attn} . 5-seed.
5. Run lever D (m_{max} sweep) single-seed at the chosen A+B+C combination. 5-seed.

Each step gates the next: if the smoke fails, halt that branch and move to the next lever in the matrix.

4. Smoke and promotion gates

Smoke gate (single-seed, abbreviated config)

For each candidate lever:

- $\text{AUC} \geq \text{current best} - 0.01$ (allow for single-seed noise).
- Wall time $\leq 2 \times$ baseline at the matching config.
- No regression in macro- F_1 below 0.50.

Pass $\Rightarrow$ promote to 5-seed paired.

Promotion gate (5-seed paired, production config)

Same protocol as the in-flight degree-adaptive 5-seed:

- Paired-mean AUC $\geq$ current best with $\sigma \geq 1.0$.
- Paired σ within $\pm 1.0\sigma$ of baseline.
- Per-seed delta consistent (no all-five-seed swings).

5. Affected files / artifacts

- `hymeko_graph/examples/probe_per_vertex_thresholds.rs` extended with c CLI arg.
- `reports/2026-05-10-epinions-lift-studies.md` this plan’s accompanying study report (CPU-only c -sweep results + next-experiment recommendations).
- `reports/figures/lift_studies_plot.py` matplotlib visualization of the cycle-distribution sweep.
- `signedkan_wip/experiments/run.*_2026_05_10.sh` orchestration scripts, one per lever; written only after the smoke gate of the prior step passes.

6. CORE.YAML touched

None. All edits in `hymeko_graph` (not core), `signedkan_wip` (not in CORE.YAML), and the existing `reports/figures/` directory. No new dependencies.

7. Halt conditions

- In-flight 5-seed shows adaptive $c=1$ regression ≥ 0.01 AUC at production config $\Rightarrow$ this whole sweep is quality-poisoned; revert and re-evaluate the smoke-result interpretation.
- Any single-seed smoke shows AUC drop > 0.05 vs current best $\Rightarrow$ halt that branch; do not 5-seed.
- GPU contention with in-flight 5-seed $\Rightarrow$ defer GPU smokes until the 5-seed completes.

8. Architecture diagram

See `plan.tikz` (lever stack) and `plan.mmd` (experiment order).

